MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF MECHATRONICS ENGINEERING

VII SEMESTER B. TECH MECHATRONICS ENGINEERING

END SEMESTER EXAM

Subject: Machine Learning **Subject Code: MTE 4073**

Time: 2:00 to 5:00 PM **Date: 17/11/2022**

Important Note: Important Note: Make necessary assumption, if required

S. No.	Question	M	CO	PO	LO	BL
1A	Explain multiple linear regression with explaining the criteria for choosing regression line. How to learn parameters for 2-d problem in regression.	5	CO2	2,5	2,3	5
1B	Define the term machine learning, explain the supervised and unsupervised machine learning with example.	3	CO1	1,2	1,2	5
1C	Compare between traditional computing and artificial neural network.	2	CO4	3,4	3,4	4
2A	For Naïve Bayes Classifier (i) Estimate conditional probabilities of each attribute {color, legs, height, smelly} for the species classes: {M, H} using the data given in the Table 4a. (ii) using these probabilities estimate the probability values for the new instance-(color=green, legs=2, height=tall, and smelly=no)	5	CO3	3,5	3,6	5

Table 4a

S. No.	Color	Legs	Height	Smelly	Species
1	White	3	Short	Yes	M
2	Green	2	Tall	No	M
3	Green	3	Short	Yes	M
4	White	3	Short	Yes	M
5	Green	2	Short	No	H
6	White	2	Tall	No	H
7	White	2	Tall	No	H
8	White	2	Short	Yes	H

2B	Define confusion matrix? Deduct the formula for accuracy using a confusion matrix.	**3**	**CO2**	2,3	2,3	5
2C	List any four major advantages of neural network.	**2**	**CO4**	2,3	2,3	4
3A	Elaborate the incremental approach to evolutionary robotics for demining and underwater exploration.	**4**	**CO4**	4,5	3,4	6
3B	Explain the cost function; how can it be minimised?	**3**	**CO2**	2,3	2,5	5
3C	Explain following with the help of examples (i) Semi Supervised learning (ii) Reinforcement learning	**3**	**CO2**	1,2	1,2	5
4A	Explain in detail for decision tree (i) Entropy (ii) Information gain (iii) Prepruning (iv) Postpruning	**4**	**CO2**	2,3	2,5	5
4B	Explain and compare the basic components of a human biological neuron and artificial neuron with the help of neat sketch.	**3**	**CO4**	2,3	1,4	5
4C	Explain single layer feedforward neural network and multilayer neural network architecture with neat sketch.	**3**	**CO4**	1,2	1,2	5
5A	Propose and elaborate the application of K-Nearest Neighbour (KNN) algorithm for human action recognition.	**4**	**CO4**	3,5	3,6	6
5B	Determine the support vectors and estimate the hyperplane using Linear SVM for positively and negatively labelled data points $\{\binom{3}{1}, \binom{3}{-1}, \binom{6}{1}, \binom{6}{-1}\}$ and $\{\binom{1}{0}, \binom{0}{1}, \binom{0}{-1}, \binom{-1}{0}\}$ respectively.	**4**	**CO3**	2,3	3,6	5
5C	Formulate following activation functions used in artificial neural networks (i) Bipolar sigmoid (ii) Hyperbolic tangent	**2**	**CO4**	2	2,3	6